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INTRODUCTION

Gradlong represents a quick method of eddy current measurement that is accurate, simple, and does not require the reconfiguration of hardware associated with past methods. Gradlong auto acquires raw data, calcs and plots B field along readout axis vs t after EC excitation pulse and decomp field into spatially invariant (B_0), linear gradient and second order terms vs. t.

This tool should be used to decide whether other more time-consuming troubleshooting methods or calibrations should be employed. This document provides instructions for running Gradlong in an independent mode. Gradlong is a very comprehensive tool. These instructions only describe the basic use of the tool.

Terminology Used For Gradlong

Donor This term refers to the **physical** gradient axis (x, y or z) *upon* which the eddy current excitation trapezoid pulse is played prior to the acquisition of data. It has been previously and somewhat nebulously referred to as the “gradient” axis in other EC measurement tools. Every gradlong measurement has an associated donor and recipient.

Recipient This term refers to the physical gradient axis (x, y or z) *along* which the eddy current related magnetic field error tail or response is measured. It has previously been referred to as the “coil” axis in other EC measurement tools. Every gradlong measurement has an associated donor and recipient.

1- PREREQUISITES TO RUNNING GRADLONG IN A STANDALONE MODE

1. The gradlong software is part of Release 9.1 and 10.x Signa service software. “Pre-Release 9.1” software, the gradlong related files, must be obtained via InSite modem ftp or separate tools CD, to be released during the second quarter of 2002.
2. A CTL sphere phantom (10 cm diameter) with foam holder placed inside the head loader shell is the preferred phantom assembly for gradlong. If no CTL phantom and altered foam fixture are available, the next preferred is the TLT sphere and loader in the head coil. The DQA phantom can be used, but only if no other phantom is available.

Note

Obtain another foam holder (2170481) and trim it down to size until it fits snugly in the bottom of the head loader shell.

1-1 Tools Required

WARNING!

POISON HAZARD! THE PHANTOM CONTAINS NICKEL, A SUSPECT CARCINOGEN. DO NOT INGEST. DISPOSE OF AS A HAZARDOUS WASTE ACCORDING TO STATE AND FEDERAL REGULATIONS.

Item	Description	Part Number	Qty.
1.	100-mm Sphere Phantom	46-317586G1	1
2.	EPI Foam Positioner	2170481	1
3.	Head Loader Shell	46-287899G1	1
4.	Head TLT Sphere (if 100-mm sphere not available)	46-265826G6	1

2- READING ON LINE HELP FOR GRADLONG TOOL

The Gradlong tool incorporates a command line Help file that can be called up for instructions on setup and switch settings. A copy of the contents of the help file appears in Appendix A of this document. To access the Help file on Signa:

1. At the host computer, go to the Service Desktop and open a "C" shell. At the command line type:

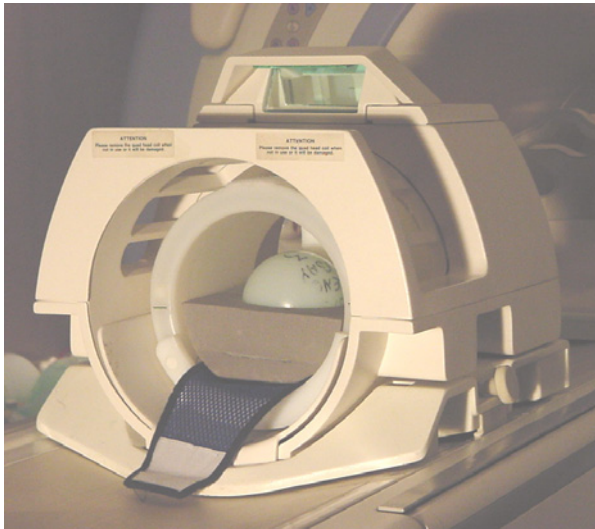
```
cd /usr/g/service/cclass <Enter>
```

```
gradlong h | more <Enter>. (Note that the | is a pipe symbol and not a colon.)
```

2. The Help file will scroll across the screen and then stop. Use the C-shell scroll bar at the left of the window to scroll through the text.

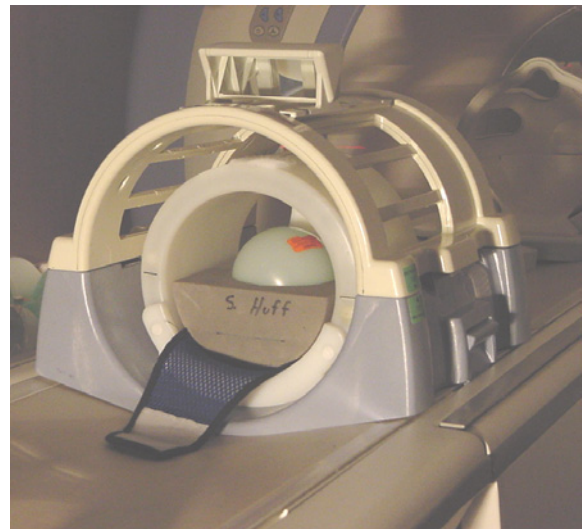
3- BASIC SCAN TO VERIFY SYSTEM EDDY CURRENT COMPENSATION

1. This check will use the head coil, CTL sphere phantom (10-cm diameter), foam holder and loader shell. Or, use the TLT phantom and loader, if available. The DQA phantom can be used but the accuracy of the results may be in question due to the structures inside the DQA phantom.
2. At the host computer, go to the Scan Desktop and select **[New Patient]**. In the Patient ID field enter **geservice** and press the **<Enter>** key on the keyboard.
3. Put the CTL sphere with foam holder and loader in the head coil. See Illustration 3-1 for the preferred setup, or Illustration 3-2 if the preferred setup is not possible. Landmark S/I center of the sphere and advance to scan.



STANDARD QUAD HEAD COIL
WITH CTL PHANTOM AND FOAM HOLDER

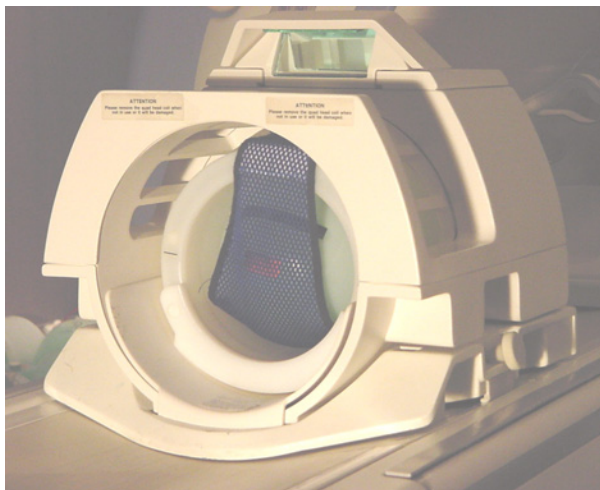
NOTE THAT STRAP IS DOWN



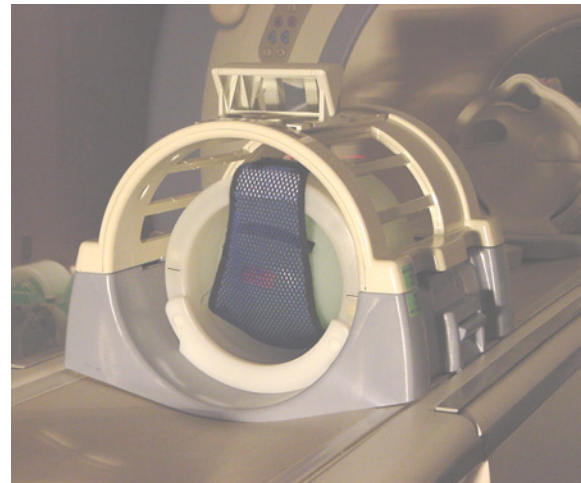
SPLIT HEAD COIL
WITH CTL PHANTOM AND FOAM HOLDER

NOTE THAT STRAP IS DOWN

PREFERRED SETUP
ILLUSTRATION 3-1



STANDARD QUAD HEAD COIL
PHANTOM IN THE HEAD LOADER



SPLIT HEAD COIL
PHANTOM IN THE HEAD LOADER

USE IF PREFERRED SETUP (ILLUSTRATION ABOVE) IS NOT POSSIBLE
ILLUSTRATION 3-2

Note

The phantom must be centered in the bore to ensure test results are accurate.

4. At the host computer, go to the Service Desktop and open a C-shell. Do **not** run this from the Xterm C-shell on the Common Service Desktop: the Xterm C-shell does not offer enough scroll (buffer) area.

- At the command prompt type the following:

```
cd /usr/g/service/cclass <Enter>
```

To start the test, type: **gradlong headsphere fpref=xxxx <Enter>**

In “fpref=”, replace the **xxxx** with a preferred prefix for easy identification of the result file later. A good choice would be to use the date; for example:

```
gradlong headsphere fpref=march21 <Enter>
```

The resulting output file will now be “/usr/g/service/cclass/march21.gradlong.tar.gz”.

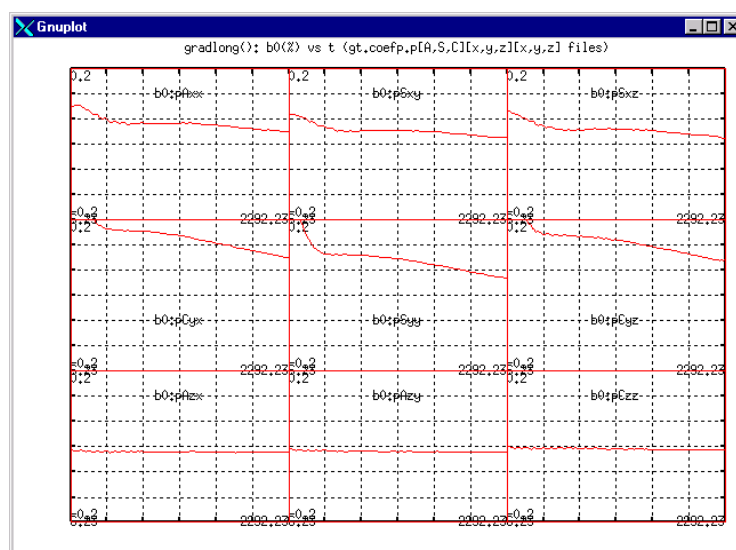
For TwinSpeed systems, the command above defaults to “Whole” mode. If you want to test “Zoom” eddy currents, the command string would then be:

```
gradlong headsphere zoom fpref=xxxx <Enter>
```

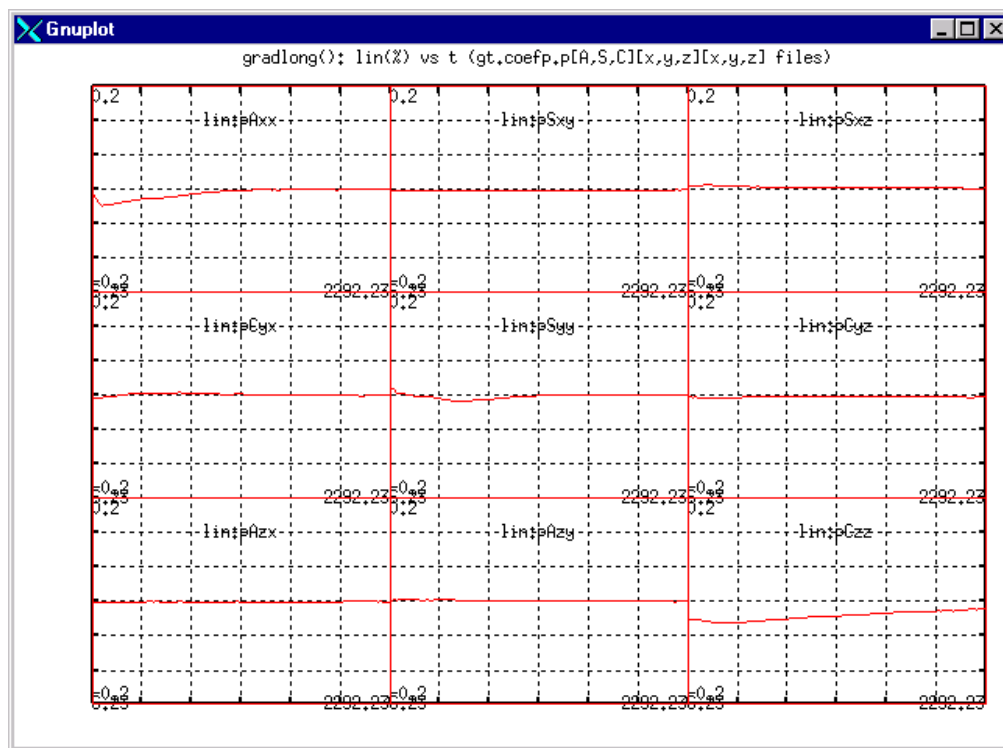
gradlong headsphere t=zz,yy,xx fpref=xxxx <Enter> measures only “on-axis” cases, WHOLE mode.

gradlong headsphere t=xx fpref=xxxx <Enter> measures one case (x as shown here), WHOLE mode.

- The resulting scan will test all axes and will last for about 15 minutes. Windows will open and close and text will scroll in the open C-shell during the operation. Do not attempt to interact with the process in the C-shell. At the completion of the scanning, test results will be shown automatically.
- At the end of the gradlong test, two plot matrices will automatically be displayed showing the eddy current-generated gradient field. Also shown are the B_0 residuals, which are the undesirable tails that decay away after the decay ramp of the eddy current excitation trapezoid pulse. Two graphs will be displayed, as shown in Illustrations 3-3 and 3-4.



B_0 PLOTS
ILLUSTRATION 3-3



LINEAR PLOTS
ILLUSTRATION 3-4

Note that each donor-recipient measurement has a constant location in the 3 x 3 display matrix. The cell annotation is either **lin** (i.e., linear gradient error), or **b₀** (i.e., B₀ field error) followed by a scan plane code (A=axial, S=sagittal, C=coronal), followed by the donor-recipient code. If a particular measurement has not been made for the gradlong run being examined, there will not be a data plot in that cell and that cell will be labeled "No_File."

Vertical axes in each linear gradient plot cell are in percentage of the eddy current excitation pulse amplitude. Vertical axes in each B₀ field plot cell are in percentage of the eddy current excitation pulse field that would be produced 1.0 cm from isocenter along the donor axis.

Horizontal axes for both linear gradient and B₀ field plots are in msec after the end of EC excitation pulse flat-top. See Table 3-1 for the display matrices map. Ideally, the output plot would be flat horizontally. Press **<Enter>** while the cursor is in the *original* shell window to erase the plots.

TABLE 3-1
DISPLAY MATRICES

	Recipient X Column	Recipient Y Column	Recipient Z Column
Donor X Row →	XX	XY	XZ
Donor Y Row →	YX	YY	YZ
Donor Z Row →	ZX	ZY	ZZ

- Vertical axes in percentage of EC excitation pulse amplitude
- Horizontal axes in msec after the end of EC excitation pulse flat-top

In another C-shell, the Spec Limit Analysis of the data collected will be printed. See Illustration 3-5. It has columns for:

- The **response** (“lin” = linear gradient, “B₀”)
- The **type** of measurement gradlong made (long or vlong)
- The **donor-recipient** axis code in the time order acquired (xx, zy, etc.)
- The **tmin(ms)** and **tmax(ms)** over which a specification has historically been set
- The **usl(%)** or upper spec limit
- The **max(%)** found over all points gradlong has measured *within the indicated time interval*, and
- The **diff**, which is the value [max(%) - usl(%)]

Note that the **diff** value is negative when the measured **max(%)** is *below* the **usl(%)**. Negative values in the **diff** column indicate that gradlong’s measure for the line is IN spec. Positive values in the **diff** column indicate that gradlong believes the measure is OUT of spec.

From file gradlong.sum:

resp	type	dr	tmin(ms)	tmax(ms)	usl(%)	max(%)	diff
lin	long	zz	2.5	10.0	0.090	0.009	-0.081
lin	long	zz	10.0	100.0	0.030	0.013	-0.017
lin	long	zz	100.0	2000.0	0.018	0.015	-0.003
Bo	long	zz	2.5	10.0	0.100	0.012	-0.088
Bo	long	zz	10.0	100.0	0.100	0.009	-0.091
Bo	long	zz	100.0	2000.0	0.100	0.021	-0.079
lin	long	yz	2.5	10.0	0.090	0.003	-0.087
lin	long	yz	10.0	100.0	0.030	0.003	-0.027
lin	long	yz	100.0	2000.0	0.018	0.004	-0.014
Bo	long	yz	2.5	10.0	0.100	0.186	0.086
Bo	long	yz	10.0	100.0	0.100	0.184	0.084
Bo	long	yz	100.0	2000.0	0.100	0.147	0.047

Annotations:

- Response:** Points to the 'resp' column.
- Type:** Points to the 'type' column.
- Donor recipient pair:** Points to the 'dr' column.
- Time Range:** Points to the 'tmin(ms)' and 'tmax(ms)' columns.
- Upper Spec Limit:** Points to the 'usl(%)' column.
- Max Read during Scan:** Points to the 'max(%)' column.
- Difference between upper spec limit and actual found during scan:** Points to the 'diff' column. A negative number here indicates that this axis is below spec and needs no further attention. Having a positive difference indicates that axis is out of spec. You'll need to re-run Grafidy for that axis.

SPEC LIMIT ANALYSIS
ILLUSTRATION 3-5

4- RESULT FILE MANAGMENT

Once a gradlong scan is completed, the results files are created and stored in **/usr/g/service/cclass**. Result files are overwritten with each pass of gradlong. The tar.gz files are not overwritten unless the same prefix is specified in the “fpref=” argument. Display management commands work on whatever un-tarred result file is open.

- Type **gradlong mb0** to display all measured **b0** residual field plots in percent vs sample time in msec, referenced from the *end* of the EC excitation pulse’s flat-top.
- If you want to change the vertical y-scale of these plots, then type “**gradlong mb0 {yscale}**” where the new vertical scale be +/-yscale in percent (for example “**gradlong mb0 1.0**”).
- Type **gradlong mlin** to display all measured linear gradient residual plots in percent vs. sample time in msec, referenced from the *end* of the EC excitation pulse’s flat-top.
- If you want to change the vertical y-scale of the these plots, then type “gradlong mlin {yscale}” where the new vertical scale be +/-y-scale in percent (for example **gradlong mlin 1.0**).
- A complete gradlong run summary can be found in the text file **/usr/g/service/cclass/gradlong.sum** (check it out using “more gradlong.sum”, “cat gradlong.sum” or your favorite text editor).
- Type **gradlong show** to display the EC calibration-related files, links and contents. This same output can also be piped to a text file by typing “**gradlong show > yourfilename**” for later transfer or use elsewhere, etc. Note that this same EC cal file information is also contained in the summary file “gradlong.sum,” which is created each time a gradlong measure run is made.
- Results for each measurement “run” are automatically stored in a zipped tar file **/usr/g/service/cclass/ [prefix.]gradlong.tar.gz** where “prefix” may have been specified via the “fpref=prefix” switch when the gradlong measurement run was specified. The default prefix is a “null” or empty string. If you specify a prefix, then a period (“.”) will automatically be added between it and the word “gradlong” in the filename. A results tar file will *always* be created regardless of whether you specify a prefix or not. If you do NOT specify a prefix, the results tar file will be named “gradlong.tar.gz”. A gradlong measurement run will ONLY automatically delete a tar file of the same name it is about to create for the current measurement. Therefore, using unique prefixes can ensure that all measurement results can be retained for later examination. The prefixes “spt”, “spt_WHOLE” and “spt_ZOOM” should not be used when running gradlong standalone since these are already used when gradlong is run via SPT, and we do want to retain the latest versions of these results files from SPT. The SPT-generated gradlong results tar files can be unzipped and untarred just like any other. DO NOT delete the files [spt,spt_WHOLE, spt_ZOOM].gradlong.tar*.
- Perform the following to examine results of a previously-saved gradlong run (i.e. *not* the immediately current run or just completed run but one “saved” by using the fpref=? switch):

1. Type **ls -l *gradlong.tar*** to list the files you have to choose from in /usr/g/service/cclass.
2. Type **gradlong clean** to delete all *generically-named* results files for the presently viewable data so these files will not be mixed or confused with data files coming from the next tar operation that unpeels a saved file. **This “clean” step is crucial.**
3. Type **gunzip [prefix.]gradlong.tar.gz** to unzip the tar file of interest. If this “unzip” operation has been previously performed on this file, go directly to the next step (i.e., the zipped file “*.gz” no longer exists after the first “unzip” operation on it).
4. Type **tar -xvf [prefix.]gradlong.tar** to peel apart the tar file into *fresh* generically-named results files for the measurement run you now want to examine. This tar operation will list the files that are contained in and coming from the tar file.

Now you’re ready to use gradlong’s “mlin”, “mb₀”, “spec”, “show”, etc. switches to redisplay the plots measurement table and text information.

APPENDIX A GRADLONG HELP FILE

gradlong <dqa, headsphere or body> [t=?] [zoom] [fpref=?] [cdpw=?] [vl] [a=?] [cp]
[(b,g,s,a)t] [apssep] [gampoff] [noplot] [d]

dqa, headsphere, body	Selects acquisition mode, NOT optional (rcv coil & phantom combo) dqa selects head coil, 24 cm FOV, for DQA phantom (don't use dqa w/HFO) headsphere selects head coil, 24 cm FOV, for 10cm or 17cm sphere. body selects body coil, 32 cm FOV, for 27cm sphere (worst choice).
t=?	Specs 2 letter keyword or list of comma separated keywords that define tests where 1st letter is physical axis donor (x,y,z,a=xy&z), 2nd letter is physical axis recipient (x,y,z,a=z,y & x or o=ON axis only) Keyword opts: [x,y,z,a][x,y,z,a] OR ao (e.g. xy,xx,az,za,zy,aa,ao..) "ao" means (a)ll donors with (o)nly on-axis recips (i.e. ao="zz,yy,xx"). Keywords with "a" for donor and/or recip must be used ALONE. Don't repeat keywords. Valid examples (head or body mode): t=xx t=ya t=xx,zx,yz t=ao t=aa t=xx,yy,zz,zx,zy,yz Order spec'd is order run ("a" orders zyx in time). In dqa mode without the apssep option there are some limitations since the first (=only) autoprescan must be done on the sagittal plane to prevent later over-ranges. In this case if ANY test has a z recipient, then the first test must have a z recipient (w/dqa mode). The default is "aa" (= "zz,zy,zx,yz,yy,yx,xz,xy,xx") if t=? switch is NOT used.
zoom	Selects "ZOOM" mode for TRM via protocol (default is "WHOLE" mode)
fpref=?	Specs prefix appended to results tarfile (ala ?.gradlong.tar.gz)
cdpw=?	Changes default EC excitation flat-top width (ms) [400,2000,6000] (def=6000)
vl (this is a v and lower case L)	Changes time mode from long (default) to VERY LONG time constant (increases TR).
a=?	Specs #acqs time interleaved for final waveforms,[1..8] (default = 1)
cp	Change default scan plane(s) used for measurements (see author for details). This switch is NOT compatible with dqa acquisition mode.
(b,g,s,a)t Self test mode	Self test mode: bt=B0 sim(w/CF increment), gt=grad sim(w/daq timeshift), st=grad sim(w/gradshim offset), at= (bt,gt,st) for first 3 tests respectively
apssep	Do separate APS per scan (defaults is 1st APS used for rest).
gampoff	Turns off (zeros amplitude) of EC grad excitation pulse (%calcs still done w/def amp).
noplot	Turns off automated results plotting at end & extends post ADVANCE_TOSC sleep time.
d	Turns debug mode ON.

NOTE: DATA MANAGEMENT & DISPLAY SWITCHES BELOW ARE USED ONE AT A TIME!

gradlong [h or help] [mlin [%]] [mb0 [%]] [show] [spec] [rpd] [ecbackup] [clean] [list]
 [mp"key" [plotoption] [p,P]] [get host] [gettar host [prefix]]

help or h	Display this help menu.
mlin [%]	Plot all available linear gradient (from +/-%), (default =0.2).
mb0 [%]	Plot all available b0 data (from +/-%), (default =0.2).
show	Display EC comp related files.
spec	List the spec comparison data from file gradlong.sum.
rpd	Resets SIGNA protocol directory to /usr/g/protocols
ecbackup	Backup SIGNA dir /usr/g/caldir EC related contents to tar file ecbackup.tar ONLY if tar file DOES NOT already exist.
clean	Cleanup data files (use before un-zipping/un-taring gradlong tar file)
list	List output files present.
mp[key] [plotoption] [p,P]	key=(plane)(donor)(recip)[.acq#]=(a,s,c)(x,y,z)(x,y,z)[.?.], like axx,syz,czx.1 key=blank= display data menu plotoption (1-15,def=blank= display plot option menu) p=+hrdcpy, P=hardcpy only (use only on workstation w/printer)
get host	Get gt.* results files from host.
gettar host [prefix]	Get [prefix.]gradlong.tar.gz file from /usr/g/service/cclass on host

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A		Hawthorne	
0	April 15, 2002	Hawthorne	Initial Release
1	Jan. 20, 2002	C. MacDonald	Changed doc title in header to SYSTSC19. Text edits and additions per Don Thomé